

1. User Profile

This study included 8 Bairn users (P1–P8). The participants differed in their baby’s age, parenting experience, and frequency of app use.

Participant	Background	Usage
P1	Baby around 1.5 years old; first-time mother	Daily use
P2	Baby over 1 year old; father; works full-time	2–3 times/day
P3	2-month-old baby; first-time mother	8–10 times/day
P4	19-month-old baby	4–5 times/day
P5	14-month-old baby; father	3–4 times/week
P6	Baby born in May of the same year	3–4 times/day
P7	3-month-old baby; first-time parent with limited experience	Approx. 2–9 times/day; varies by situation
P8	6-month-old baby; new mother with limited systematic parenting knowledge	6–7 times/day

Key User Characteristics:

Different developmental stages: babies ranged from 2 to 19 months old.

Different parenting experience: some participants were explicitly identified as first-time parents or as having limited parenting experience.

Different usage frequency: participants included both high-frequency and relatively low-frequency users.

Different primary needs: daily childcare recording, parenting knowledge and guidance, and child growth and health information.

Overall, participants differed mainly in their baby’s developmental stage, parenting experience, usage frequency, and specific needs. These differences are further explored in the following themes.

Theme 1: Integrating Bairn into Everyday Parenting Activities and

Usage Needs

2.1 Corresponding Axial Coding

Recording behaviours during daily childcare

Integrating digital recording into daily parenting routines
Digital records supporting parenting information management
Digital records supporting understanding of the baby's daily condition
Parenting knowledge needs and childcare uncertainty
Parenting contexts influencing app use

2.2 Theme Overview

This theme explores why parents used Bairn, how Bairn was integrated into their everyday parenting activities, and the practical role of digital records in daily childcare.

The interviews showed that parents' needs for using Bairn were not limited to a single function. On the one hand, parents needed to record and manage a large amount of information generated through daily childcare, including feeding, sleep, nappies, pumping, and growth and development. On the other hand, some parents, particularly those with limited parenting experience, used Bairn because they lacked systematic parenting knowledge, wanted professional guidance, and experienced uncertainty about their baby's condition. This suggests that parents' needs for using Bairn came both from information management during everyday childcare and from the knowledge and support needs that arose during parenting.

Further analysis showed that parents did not treat Bairn as an app completely separate from real-life childcare, but integrated it with their existing caregiving activities. Everyday activities such as feeding, sleep, and nappy changes directly generated recording needs, while parents also decided when to record according to the timing, childcare arrangements, and their individual circumstances.

At the same time, the value of digital records was not simply to 'save information'. Some parents mainly relied on personal memory, observation, or handwritten notes before using Bairn, while digital records allowed this information to be stored externally and reviewed later. Parents also used historical records to understand their baby's daily routines, confirm completed childcare activities, and observe changes in their baby's development.

Therefore, the core research finding of this theme is that parents used Bairn to address multiple needs in everyday parenting, including managing continuously generated childcare information, compensating for limitations of personal memory, and seeking support when they had limited parenting experience or faced uncertainty in childcare. Bairn could be integrated into everyday parenting activities because digital recording was combined with existing caregiving routines, allowing parenting information to be continuously stored, reviewed, and used. At the same time, parents' specific ways of using the app varied according to the baby's developmental stage, childcare needs, and individual parenting contexts.

2.3 Research Finding 1: Daily Childcare Activities as Key Triggers for Bairn Use

The interviews showed that parents' use of Bairn was not completely independent of their

everyday parenting activities. Activities such as feeding, sleeping, changing nappies, and pumping directly generated recording needs and became important contexts for opening the app.

Therefore, Bairn use should not simply be understood in terms of how many times parents opened the app each day, but rather in terms of how digital recording was connected to real-life parenting activities.

Supporting Evidence:

Interpretation:

Participant	Open Code	Original Interview Evidence
P1	Recording activities as part of the daily routine	I can simply open the app whenever I'm looking after my baby and record activities naturally as part of my daily routine.
P2	Records feeding immediately after feeding	Every time I finish feeding my baby, I open the app and enter the feeding type and the feeding amount.
P3	Open app during feeding / sleep / nappy changes	I usually open it when feeding my baby, when my baby is sleeping, or when changing nappies.
P6	Opens the app after completing childcare tasks	Basically, I open it after completing a childcare task, such as feeding, changing the baby's nappy, or putting the baby to sleep.
P7	Records feeding information immediately / Enters records in batches at night	During the day, I immediately recorded the amount of milk after feeding or expressing breast milk; at night, after putting my baby to sleep, I entered the day's sleep and bowel movement records all at once.

These findings show that Bairn use is not an activity separate from everyday parenting, but a recording process that is repeatedly triggered by actual childcare activities such as feeding, sleeping, nappy changes, and pumping. Parents usually record after completing the relevant activity, but the timing of recording is adjusted according to their childcare arrangements and available time.

For example, P7 records feeding and pumping information immediately during the day, while entering other records after the baby falls asleep at night. This shows that parents do not necessarily record mechanically after every activity, but adjust their recording practices according to the actual parenting context.

Therefore, one important reason why Bairn can be integrated into everyday parenting is that digital recording can be attached to existing childcare activities, rather than requiring parents to create an additional and separate routine for using the app.

2.4 Research Finding 2: Limited Parenting Experience and Caregiving Uncertainty Motivate Some Parents to Seek Support from Bairn

Parents' reasons for using Bairn were not limited to recording daily activities. For some participants, the need to use the app also arose from limited parenting experience, a lack of systematic parenting knowledge, and uncertainty when dealing with specific situations involving their babies. This suggests that, for some parents, Bairn served not only as an information-recording tool but also as a way of seeking external support during parenting.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P3	First-time parent; Learning childcare skills through app	"Since I'm a first-time mum, I don't know much about parenting. I learned a lot from the NHS Guides, such as the correct feeding methods, and I followed the app's guidance when feeding my baby."
P5	Started using the app to learn about parenting; Wanted to learn more about parenting	"When I first started using the app, I wanted to learn more about parenting and keep a good record of my child's growth. I hoped the app would help my baby grow in a healthier way."
P7	Limited parenting experience; Seeking professional guidance	"My parenting experience is quite limited, and I wanted to get some professional guidance and advice."
P8	Lack of systematic parenting knowledge; Limited parenting guidance from family and friends; Bairn as parenting reference tool	"I mainly started using it because I lacked a systematic understanding of parenting knowledge. I often felt overwhelmed, and the effective parenting guidance I could get from family and friends was quite limited."

Interpretation:

These findings show that parents' need to use Bairn was related to their parenting stage and prior experience. For parents with limited parenting experience, everyday childcare was not necessarily a familiar and fixed routine, but an ongoing process in which they encountered new questions and needed to confirm and learn. Therefore, they used Bairn not only to record what had already happened, but also to obtain knowledge and support through the app.

It is also important to note that high-frequency use does not necessarily mean that parents simply liked the app more. It may also indicate that parents were facing stronger information needs and greater uncertainty in their current parenting situation. Therefore, understanding why parents use Bairn requires consideration of their parenting experience and the specific childcare

problems they are facing, rather than simply comparing the frequency of use.

2.5 Research Finding 3: Digital Records as External Support for Compensating for Limitations of Personal Memory

Some parents mainly relied on personal memory, observation, or handwritten notes to manage their baby's daily information before using Bairn. However, everyday childcare involves a large amount of repetitive information, making personal memory alone unreliable. Digital records therefore served a function beyond simply "storing data."

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P1	Previously forgetting feeding times; Previously forgetting sleep times; Uses the app as an external memory aid	"I used to forget when my baby last slept or had milk. With this app, everything can be recorded, including the exact amount of milk my baby drinks each time."
P2	Convenient access to historical records; Recording feeding information; Comparing feeding records	"Every time I finish feeding my baby, I open the app and enter the feeding type and the feeding amount."
P6	Memory is not always reliable for childcare tracking; App is more reliable than memory; Uses app instead of relying on memory	"My memory isn't very good, and when I'm at work, I sometimes forget to record things." / "For example, recording feeding times is definitely more reliable than relying on my own memory."

Interpretation:

What is important here is not simply that parents "liked recording," but that digital records reduced the need for parents to continuously remember large amounts of repetitive childcare information. For example, parents did not need to constantly remember when their baby had been fed or slept, because this information could be stored in the app and reviewed later. Therefore, Bairn's recording function effectively created a form of external memory support: personal memory → digital recording → information storage → later review. This suggests that the value of digital recording comes from responding to a practical information-management problem in everyday childcare, rather than simply from having a recording function.

2.6 Research Finding 4: Historical Records Enable Parents to Further Understand and Manage Their Baby's Daily Condition

Digital records do not lose their value once information has been entered. Some parents revisited previously stored feeding, sleep, meal, and growth information to understand their baby's daily patterns, confirm their current caregiving status, or observe changes over time.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P1	Reviewing previous sleep records; Checking previous routines; Using records to recall previous childcare activities	"The most recent time was yesterday. I opened the feeding records, logged my baby's feeding, and then checked the previous sleep records."
P2	Reviewing previous feeding records; Comparing feeding records; Checking daily feeding frequency; Checking daily feeding amount	"In the feeding records, I could see the feeding type and the amount of feeding, and I compared this record with the previous ones."
P6	Checking whether baby has already been fed; Checking nappy changes; Coordinating caregiving activities	"If two people are using the app on two different devices and the records can be shared—for example, showing how many feeds the baby has had today or how many nappies have been changed—that would definitely be much better."
P7	Tracking baby's growth information; Recording weight and height; Reviewing growth information	"Occasionally I use: vaccine & health check reminders, weight, height and head circumference growth curve records; I open it to check the relevant information before health checks and vaccinations."
P4	Monitoring long-term growth changes; Following developmental changes over time; Continues recording height and weight	"The core purpose of logging is to observe long-term growth changes, so there's no need to record repetitive, regular data. I only keep logging metrics like height and weight that reflect long-term changes."

Interpretation:

These materials show that digital records are not one-time “information inputs.” Parents can use previously stored information through the process of recording → reviewing/comparing → understanding the current situation → supporting childcare management. For example, historical feeding information can help parents understand the baby's feeding situation, growth records can help them observe long-term changes, and completed childcare records can help them confirm the baby's current status. Therefore, Bairn's value extends beyond simply “storing information” to using stored information for everyday parenting management.

2.7 Research Finding 5: Parents' Ways of Using Bairn Are Influenced by the Baby's Developmental Stage and Specific Parenting Contexts

Participants differed in their frequency and ways of using Bairn, but these differences cannot simply be attributed to being “high-frequency” or “low-frequency” users. Further analysis

showed that the baby's developmental stage, current childcare needs, and parents' own life arrangements influenced which functions parents used and how frequently they used the app.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P3	Young baby; High-frequency childcare recording needs	"Because my baby is still very young and needs frequent feeding and sleep, so it's important to keep records."
P7	Developmental stage affects recording focus; High-frequency feeding needs	"My 3-month-old baby is mainly breastfed, so pumping, storing breast milk, and managing the milk supply are the most frequent tasks in my daily routine. It's quite time-consuming, so this feature of the app is really useful for me."
P4	Older baby; Reduced repetitive recording; Focus on long-term growth	"For older babies like my 19-month-old son, we no longer need to record milk intake since milk is just an occasional drink."
P2	Work schedule affects usage time; Evening app use	"Usually after work when I have some free time. I mostly use it in the evening, and I rarely open it during the day because I'm at work."
P6	Actual childcare activities influence use	"Basically, I open it after completing a childcare task, such as feeding, changing the baby's nappy, or putting the baby to sleep."

Interpretation:

These materials show that parents' ways of using Bairn are not fixed, but are shaped by the baby's developmental stage, actual childcare needs, and parents' life contexts. Younger babies require more frequent feeding and sleep-related care, creating greater recording needs; as babies grow, the focus of recording also changes. Parents' work and daily arrangements further influence when they use the app.

2.8 Theme-level Interpretation: Overall Interpretation of the Theme

Overall, parents used Bairn not only to record information about their babies, but also to respond to information-management, memory-support, and parenting-knowledge needs in everyday childcare.

Digital records could be integrated into daily care activities and help parents store, review, and use childcare information. At the same time, the baby's developmental stage, parents' parenting experience, and specific life contexts influenced how Bairn was used.

Therefore, Bairn provides information-management and memory support by integrating digital recording into everyday parenting, while its use changes according to the parenting context of both the baby and the parent.

Theme 2: Bairn Provides Support for Parents in Managing Parenting Uncertainty

3.1 Corresponding Axial Coding

- Parenting knowledge needs and early parenting uncertainty
- Seeking reliable parenting information and guidance
- Transforming parenting knowledge into practical childcare actions
- Information understanding and actionability
- Uncertainty about health and development information
- Matching information needs with the baby's developmental stage
- Relationship between digital parenting support and professional support

3.2 Theme Overview

This theme explores why parents need information and guidance from Bairn and how this support helps them deal with limited parenting knowledge and uncertainty in childcare.

The interviews showed that some parents, particularly those with limited parenting experience, used Bairn to obtain parenting knowledge and professional guidance. For example, P7 wanted professional parenting advice, P8 lacked systematic parenting knowledge, and P3 used the NHS Guides to learn about appropriate feeding practices.

Further analysis showed that parents did not simply need more information. They needed information that could be understood and applied in actual childcare. They wanted feeding, health, and developmental guidance that was relevant to their baby's current stage, with images, videos, or clear steps helping them understand and carry out specific actions.

Uncertainty was particularly evident in health-related topics such as stool colour, Growth Charts, percentiles, and CPR. Parents needed not only to know what these things were, but also to understand whether they were normal, what they meant, and what they should do next.

Therefore, the core finding of this theme is: Parents used Bairn not simply to obtain information, but to seek support that could reduce parenting uncertainty and help them take practical childcare actions. The value of this support depended on whether the information was understandable, relevant to the baby's developmental stage, and actionable.

3.3 Research Finding 1: Limited Parenting Experience and Limited Existing Support Motivate Parents to Seek Digital Parenting Support

An important reason why parents used Bairn was their lack of sufficient experience and reference

points when dealing with early childcare tasks. The key issue was not simply that parents wanted to learn parenting knowledge, but that when they encountered unfamiliar parenting problems, their existing knowledge and social support were not always sufficient, leading them to seek external support.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P3	First-time parent; Learning parenting knowledge	"Since I'm a first-time mum, I don't know much about parenting. I learned a lot from the NHS Guides..."
P7	Limited parenting experience; Seeking professional guidance	"I have limited parenting experience, and I would like to get some professional guidance and advice."
P8	Lack of systematic parenting knowledge; Limited social support	"I lacked a systematic understanding of parenting knowledge, and I received very little effective parenting guidance from family and friends."

Interpretation:

These materials show that the need for digital parenting support did not simply come from wanting more information. It was shaped by the combination of limited parenting experience, limited existing support, and the need to make judgements about specific childcare situations. When sufficient reference points were unavailable, parents actively sought external information sources such as Bairn.

3.4 Research Finding 2: Parents Face Difficulty Transforming Parenting Knowledge into Practical Actions

Although parents could obtain parenting knowledge from Bairn, knowing something and being able to carry it out in practice were not necessarily the same.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P3	Learning feeding methods; Applying app guidance	"I learned the correct way to feed my baby and followed the recommendations provided in the app."
P5	Lack of meal-feeding guidance; Need practical guidance	"The meal-feeding section does not provide any guidance or suggestions..."
P8	Need step-by-step guidance; Need visual demonstrations	"Add short videos and step-by-step illustrations for all key actions..."
P8	Difficulty understanding text-only CPR guidance	"There were no illustrations or short videos on the page, so it was difficult to understand from the text alone, and I wasn't sure whether I was doing it"

		correctly.
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Interpretation:

These materials show that parents needed more than simply “knowing”; they needed to be able to act on what they learned. Digital parenting support therefore involves a process of transforming information into practical action, and concrete steps, illustrations, and videos can help reduce this gap.

3.5 Research Finding 3: The Presence of Information Does Not Mean That Parents Can Understand and Use It

The interviews further showed that simply providing relevant content in Bairn did not necessarily mean that parents could benefit from it. The amount of information, the way it was presented, and the use of professional terminology all influenced parents' understanding of the information.

Supporting Evidence

Participant	Open Code	Supporting Evidence
P2	Information overload; Text-heavy information; Limited visual support	"There was a lot of content, and it didn't feel very intuitive. It was all text, without any pictures or videos, so I didn't continue reading."
P3	Need for visual support; Difficulty understanding text-only guidance	"The NHS Guides are very good, but I think they would be even better if they included more pictures and illustrations."
P8	Need for visual demonstrations; Difficulty with text-only information	"It's all text, without step-by-step illustrations or videos. For practical tasks such as swaddling and burping, these are the areas that most need improvement."
P2	Difficulty understanding Growth Chart; Unclear professional information	"I wasn't sure what the different colours or the percentiles actually represented."

Interpretation

The key issue is not simply whether parents can find information, but whether they can understand and use it. Information overload, limited visual support, or insufficient explanation of professional information can prevent information from becoming practical support. Therefore, digital parenting tools need to consider the process from information availability to information understanding and use.

3.6 Research Finding 4: Uncertainty About Health and Development Increases Parents' Need for Judgement Support

Compared with general parenting knowledge, health and development-related issues involve greater uncertainty. Parents need more than an explanation of what something is; they also need to know whether it is normal, whether they should be concerned, and what they should do next.

Supporting Evidence

Participant	Open Code	Supporting Evidence
P7	Uncertainty about stool colour; Need for health explanations	"When my baby's stool colour changes, I sometimes don't know what it means or whether it is normal or a sign of illness. I would like the app to provide guidance through images and videos."
P7	Difficulty understanding Growth Chart; Unclear medical terminology	"I can't understand the icons and professional terminology in the growth chart percentiles."
P8	Need for practical CPR guidance; Difficulty with text-only information	"There were no illustrations or short videos on the page, so it was difficult to understand from the text alone, and I wasn't sure whether I was doing it correctly."
P7	Need for professional medical support; No online consultation	"There is no online medical consultation option. When I'm unsure about something, I have to use another consultation platform or go directly to the hospital."

Interpretation:

These materials show that in health and development contexts, parents' needs move from information seeking toward judgement support. They need to understand not only what something is, but also whether it is normal and what they should do. At the same time, P7 still sought professional medical support when uncertain, suggesting that Bairn is better understood as a tool for initial understanding and judgement rather than a replacement for professional medical services.

3.7 Research Finding 5: As the Baby Grows, Parents' Needs and Their Demand for Digital Parenting Support Change

Parents' needs are not static. As the baby grows, the parenting tasks parents face and the problems they need to address change, and their needs for digital information change accordingly.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P2	Need for age-specific	"I was hoping it would provide more specific

	guidance	advice for different stages of a child's development, but after browsing it, I felt that the content was quite general."
P3	Need for stage-specific feeding guidance	"I think there should be feeding recommendations that change as the baby grows."
P7	Need for age-based feature adjustment	"I would most like to see the interface customized by the baby's age, automatically hiding functions that are not relevant to the current age."
P8	Content not fully matched to the current developmental stage	"The functions are all useful, but the focus is mainly on newborns aged 0–3 months. For babies over 4 months, content on introducing complementary foods, gross motor development, and common childhood illnesses is limited."

Interpretation:

These materials show that parents' information needs change as their babies grow. Different developmental stages involve different parenting problems, so parents need information that is more closely matched to the current stage rather than simply more content. A fixed set of information may contain a lot of material while still being less relevant to parents' current needs.

3.8 Theme-level Interpretation

Overall, the findings show that parents use Bairn for parenting support through a process that moves from parenting uncertainty to practical action:

Limited parenting experience / limited existing support → specific childcare problems → uncertainty → seeking information → understanding and judging information → adapting it to the baby's developmental stage and turning it into practical childcare actions.

If information is excessive, difficult to understand, poorly matched to the baby's current stage, or lacks practical guidance, simply providing the information may not provide meaningful support.

Therefore, the core finding of this theme is that parents need not simply more parenting information, but information that can reduce uncertainty, support judgement, and guide practical childcare actions. The value of digital parenting support depends on whether information is understandable, relevant to the baby's current developmental stage, and actionable.

Theme 3: Challenges and Adaptation in Sustaining Digital

Recording in Real-World Parenting Contexts

4.1 Corresponding Axial Coding

Conflicts between parenting responsibilities and the time and attention required for digital recording

Prioritising childcare over immediate recording

Delayed recording and risk of omission due to reliance on memory

Additional operational burden during digital recording

Different recording strategies across parenting situations

Reminders and automation to reduce the burden of active recording

4.2 Theme Overview

This theme explores how parents coordinate childcare and digital recording in everyday parenting situations, and why sustained, timely, and complete recording can remain difficult even when parents recognise the value of Bairn's recording functions.

The interviews showed that parents do not record in stable or predictable situations. Activities such as feeding, nappy changing, settling the baby to sleep, and night-time care are often immediate, so parents usually prioritise caring for the baby rather than recording information. P4 explicitly prioritised childcare over recording, while P7 described caring for a newborn as requiring considerable time and energy.

When parents cannot record immediately, they may delay recording, add information afterwards, or record several items together. However, these strategies increase reliance on memory and may lead to missing information. For example, P7 sometimes added information from memory, while P6 mentioned forgetting to record sleep. Some participants also forgot to stop the feeding or sleep timer.

Recording and editing can also create additional operational burden. P3, P5, and P7 mentioned that modifying timers could be time-consuming or cumbersome. When the effort required to correct records becomes too high, parents may be less willing to continue recording.

Therefore, the core finding of this theme is: Digital recording can conflict with the time, attention, and operational demands of real-world parenting. Parents adapt through delayed or batch recording, but these strategies increase reliance on memory and the risk of missing information. Reminders and automation may therefore help reduce parents' reliance on active memory and manual input, making recording easier to integrate into everyday childcare.

4.3 Research Finding 1: Immediate Parenting Needs Take Priority Over Digital Recording

Parents need to manage childcare and recording at the same time, but when the baby has an immediate need, caring for the baby usually takes priority over recording.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
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P4	Prioritising childcare over recording; Limited time for recording	"When the baby wakes up crying, the first thing I'll do is soothe and feed him, instead of grabbing my phone to record sleep."
P7	Newborn care takes time; Difficulty recording at night	"I often don't record at night when feeding the baby because it's too troublesome."
P8	Limited time and energy for parenting	"As a mother, I'm busy every day and hardly have the energy to specifically record my mood."

Interpretation:

These materials show that missing records does not necessarily mean that parents do not value recording. Instead, immediate childcare needs and digital recording compete for the same limited time and attention. When the baby needs care, parents deal with the immediate need first and may postpone recording.

4.4 Research Finding 2: Parents Adjust the Timing of Recording According to Childcare Situations

When time is limited, parents do not simply choose between recording and not recording. Instead, they adjust when they record according to the childcare situation, either recording immediately or later.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P6	Recording after childcare activities	"I usually open it after completing a childcare task, such as feeding, changing the baby's nappy, or putting the baby to sleep."
P7	Immediate recording; Batch recording	"During the day, I record the milk amount immediately after feeding or pumping; at night, I enter the day's sleep and nappy data together."
P2	Recording immediately after feeding	"Every time I finish feeding my baby, I open the app and enter the feeding type and the feeding amount."

Interpretation:

These behaviours show that parents develop contextual recording strategies rather than following a fixed recording routine. Some information is recorded immediately, while other information is entered after childcare tasks or in batches later.

4.5 Research Finding 3: Delayed Recording Increases Reliance on Memory and the Risk of Missing Information

When parents cannot record information immediately, the task is postponed rather than removed. This means that parents need to remember the relevant information later, increasing

the possibility of missing records.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P1	Previously forgetting feeding and sleep times	"I used to forget when my baby last slept or had milk."
P6	Reliance on memory; Easily forgetting sleep records	"Sometimes I forget because once the baby is asleep, I might go off and do something else."
P7	Recording later from memory	"I sometimes add the information later based on my memory."
P3/P5/P7	Forgetting to stop the Feeding Timer	"I forgot to stop the feeding timer."

Interpretation:

This creates an important tension: digital records can reduce parents' reliance on memory, but when parents cannot record immediately, they need to remember to return and record the information later. Delayed recording therefore helps parents adapt to real childcare situations, but it can also increase memory demands and the risk of missing information.

4.6 Research Finding 4: Additional Operations in Recording and Data Processing Increase the Burden of Sustained Use

The burden of digital recording does not only come from the initial input, but also from modifying, calculating, and maintaining data. When recording requires repeated operations or additional calculations, a simple recording task can become an additional data-processing burden.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P3	Difficulty editing Feeding Timer; Repeated tapping to adjust time	"Editing the time afterwards isn't very difficult, but it would be much more convenient if I could adjust the time by sliding instead of tapping repeatedly."
P5	Time-consuming Feeding/Sleep Timer editing	"Another issue is that it is not very convenient to edit the feeding timer or the sleep timer after recording."
P5	Additional weighing and calculation for weight recording	"Few parents can weigh the baby alone directly. We usually weigh ourselves holding the baby first, then weigh ourselves separately. The app could add an automatic calculation feature for the weight difference, so I don't have to calculate manually with a calculator before inputting the number."
P7	Cumbersome editing; Repeated tapping	"Sometimes I forget to stop the feeding timer. When I try to edit the feeding duration, I have to keep tapping, and eventually I find it too troublesome and stop."

Interpretation:

These materials show that the burden of digital recording involves not only entering data, but also modifying, calculating, and maintaining it. When a simple task requires repeated taps or additional calculations, the recording cost increases and may affect parents' willingness to continue recording.

4.7 Research Finding 5: Parents Are Not Rejecting Digital Recording, but Seeking Lower-Burden Ways to Record

Although real-world parenting and recording operations can create additional burdens, parents do not completely abandon digital recording. Instead, they maintain use through different strategies and expect more reminders and automation from the app.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P1	Using reminders; Reducing forgetting	"The app also has a built-in reminder function that sends notifications to my phone at the right time."
P1	Reminders help avoid missing care times	"Since using the app's reminders, I have been able to put my baby down for naps on time."
P6	Need reminders for sleep recording	"Sometimes it's difficult to remember to use it."
P6	Need intelligent, proactive reminders	"I don't always remember to press Start, and then press Stop again when the baby wakes up."

Interpretation:

These materials show that the problem is not that parents do not want to record, but that they cannot always remember to record at the right time. They therefore want reminders and automation to reduce the burden of active memory and operation, making recording easier to integrate into everyday parenting.

4.8 Theme-level Interpretation

Overall, the findings show that parents using Bairn for digital recording need to continuously coordinate real-world childcare activities with recording requirements:

Immediate childcare needs → recording is delayed → reliance on memory for later entry → increased risk of missing information → additional burden from editing/calculation → demand for reminders and automation.

Therefore, recording difficulties are not simply caused by a lack of recording habits. They are shaped by the immediacy of childcare activities, limited time and attention, and the operational costs of recording itself.

The core finding of this theme is that digital recording needs to adapt to the fragmented and unpredictable nature of real-world parenting, rather than requiring parents to change their parenting routines to fit the app. Reminders, automation, and simpler operations can reduce the

active memory and operational burden placed on parents during recording.

Theme 4: From Individual Parenting Records to Adaptive and Collaborative Parenting Support

5.1 Corresponding Axial Coding

- Changes in parenting needs as the baby develops
- Changes in the relevance of app features across developmental stages
- Changes in recording priorities as the baby grows
- Development-stage-specific information support
- Dynamic adaptation of the app to the baby's development
- Information sharing and care coordination among multiple caregivers
- The influence of parents' work arrangements and involvement on app use
- Extending individual parenting management to family and social support

5.2 Theme Overview

This theme explores how parents' needs change as the baby grows and family caregiving situations change, and how these changes shape their needs and expectations for Bairn.

Theme 1 showed how Bairn becomes part of everyday parenting, Theme 2 explained why parents need information and guidance, and Theme 3 showed how real-world parenting situations affect sustained recording.

Further analysis showed that parents' needs for Bairn are not fixed. First, the baby's development changes the parenting tasks parents need to focus on. When babies are younger, feeding and sleep information may need to be recorded frequently. As the baby grows, some repetitive records become less important, while height, weight, and longer-term growth information become more relevant. P3 reflected the higher need for feeding records during the early stage, while P4 focused more on longer-term growth changes.

Second, shared caregiving can turn individual records into a tool for information coordination. When parents, grandparents, or other caregivers are involved, different caregivers may need to know which childcare activities have already been completed. P6, for example, mentioned using shared records to check whether the baby had already been fed and avoid duplicate feeding.

Therefore, the core finding of this theme is that parents' needs for digital parenting tools are reshaped as the baby's developmental stage and family caregiving situation change. Parents need more than a fixed set of features; they need a digital tool that can adapt to the baby's growth and support information sharing and coordination among caregivers.

5.3 Research Finding 1: Baby's Growth Changes the Information Parents Need to Track

Parents' recording needs change as their baby grows. When babies are younger, parents need to record daily care information more frequently, while the importance of some repetitive records decreases as the baby gets older.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P3	Young baby requires frequent feeding records; Need to keep daily care records for young babies	"My baby is still very young and needs frequent feeding and sleep, so it is important to keep records."
P4	Older babies no longer require milk logging; No need to record repetitive data	"There is no need to record repetitive, regular data."
P4	Continues recording height and weight; Logging is for monitoring long-term growth changes	"I only keep logging metrics like height and weight that reflect long-term changes."
P1	Features no longer relevant after one year	"I think it is helpful for parents with newborn babies, but it becomes less useful as children get older."
P7	Features not relevant to current developmental stage; Features not age-appropriate	Some features do not match the baby's current developmental stage.

Interpretation:

These findings show that the information parents need to record changes as their baby grows. In the early stage, parents mainly focus on feeding and sleep. As the baby grows, they reduce repetitive records and pay more attention to height, weight, and other long-term growth information. Therefore, app functions need to adapt to different developmental stages.

5.4 Research Finding 2: As the Baby Grows, Recording Shifts from Daily Management to Long-Term Growth Observation

The purpose of recording also changes as the baby grows. Early records mainly support daily care management, while later records are increasingly used to observe longer-term development.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P3	Young baby requires frequent feeding records; Need to keep daily care records for young babies	"My baby is still very young and needs frequent feeding and sleep."
P4	No need to record repetitive data;	"There is no need to record

	Diet and bowel habits have become routine	repetitive, regular data."
P4	Monitoring long-term growth changes; Continues recording height and weight	"The core purpose of logging is to observe long-term growth changes."
P5	Using records to keep track of child development; Following developmental changes over time	"I wanted to learn more about parenting and keep a good record of my child's growth."
P7	Reviewing growth information before health checks; Tracking baby's growth information	"I review the growth-related information before health checks."

Interpretation:

These findings show that the role of digital records changes from recording "what happened today" to observing "how the baby has changed over time." Therefore, Bairn's value is not only in storing daily information, but also in helping parents understand their baby's longer-term growth.

5.5 Research Finding 3: Shared Care Turns Individual Records into a Tool for Care Coordination

When several family members care for the baby, records are no longer used only to support one parent's memory. They also help different caregivers share information and coordinate care.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P2	Sharing childcare information with family members; Synchronising records with family members	"I would like to share and synchronise the childcare records with my family members."
P6	Multiple caregivers may feed the baby; Mother-in-law may have already fed the baby	"Other caregivers may have already fed the baby."
P6	Wants multiple people to record in the same app; Wants shared records among caregivers	"I would like multiple caregivers to be able to record in the same app."
P6	Wants to avoid duplicate feeding; Wants to confirm whether the baby has been fed	"I want to check whether the baby has already been fed so that we don't feed the baby twice."
P6	Wants shared feeding records; Wants shared nappy records	"I would like to share feeding and nappy records with the other caregiver."

Interpretation:

These findings show that when childcare is shared by multiple people, digital records extend from personal memory support to care coordination. Shared records help caregivers understand the baby's current care status and reduce information gaps and duplicated care.

5.6 Research Finding 4: Work Arrangements and Caregiving Involvement Affect the Need for Shared Information

Parents' work arrangements and family caregiving roles affect how much they can participate in daily childcare, which can increase the need for shared information.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P2	Full-time working parent; Work limits app use during daytime	"Usually after work when I have some free time. I mostly use it in the evening, and I rarely open it during the day because I'm at work."
P6	Work limits childcare involvement; Wife takes on more childcare responsibilities	"My wife is currently on maternity leave, so she naturally takes on more of the childcare responsibilities. As a father, because I'm working, I can't contribute as much to childcare as my wife does."
P6	Wants both parents to use the app and share records	"If two people are using the app on two different devices and the records can be shared... that would definitely be much better."

Interpretation:

Shared records are not only convenient for multiple users. They can also help compensate for differences in parents' availability and caregiving involvement.

5.7 Research Finding 5: Parents Expect the App to Adapt Its Support as the Baby Grows

As the baby's developmental stage changes, parents' information and feature needs also change. Parents therefore expect Bairn to adapt its content and interface to the baby's current stage.

Supporting Evidence

Participant	Open Code	Supporting Evidence
P6	Wants feature positions to adapt to baby's age; Wants features to change as the baby grows	"I think the functions should change according to the baby's age."
P7	Wants a customised interface based on baby's age	"I hope the app can customize the interface according to the baby's age."
P7	Wants irrelevant features hidden automatically	"I hope irrelevant features can be automatically hidden."
P7	Wants age-based module customization	"I hope the modules can be adjusted automatically according to the baby's age."
P2/P7	Need age-specific guidance;	"If the NHS guidance could be organised by

	Need information relevant to current developmental stage	age group, I could easily view information that is relevant to my baby."
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Interpretation:

These findings show that parents do not simply want more features. They want information and functions that are relevant to the baby's current developmental stage and that can change as the baby grows.

5.8 Supplementary Finding: Some Parents Want Social Support Beyond Individual Records

P4 suggested adding a playdate community, nearby family matching, and interaction with families with babies of similar ages. Because this need was mainly raised by P4, it is treated as a supplementary finding rather than a general need.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P4	Wants a playdate community; Wants nearby family matching	"I hope a playdate community can be added. It can match families within a 3-kilometre radius."
P4	Lack of social interaction; App focuses on private record-keeping	"This app is too inward-focused and only supports individual private records."

Interpretation:

This finding suggests that digital parenting support may extend beyond individual record-keeping to include peer interaction and social support. However, because it is mainly supported by P4, it should remain a supplementary finding.

5.9 Theme-level Interpretation

Parents' needs for Bairn change with the baby's developmental stage and the family's caregiving situation. As babies grow, recording shifts from frequent daily care information toward longer-term growth. When several caregivers are involved, individual records also become a shared basis for coordinating care.

Therefore, parents need more than a fixed set of features. They need a digital parenting tool that can adapt its information and functions as the baby grows and support information sharing among caregivers.

**Theme 5: From Available Features to Meaningful Parenting Support:
Barriers in Discovering, Understanding, and Applying Information**

6.1 Corresponding Axial Coding

Feature entry points and names affect feature discoverability

Icons and age displays affect interface understanding

Growth charts, percentiles, and medical terms increase information interpretation difficulty

Health information is difficult to directly translate into judgements about the baby's condition

Large amounts of parenting information and text-heavy presentation increase reading and filtering burden

Parents need images, illustrations, and short videos to reduce comprehension and operation costs

Practical parenting skills require step-by-step and demonstrative guidance

Continued use can improve familiarity, but cannot rely entirely on users adapting by themselves

6.2 Theme Overview

This theme examines the functional discoverability, information comprehension, and practical use barriers that parents encounter when using Bairn, and how these barriers affect the practical value of its features.

Interviews showed that some features are available, but parents may not be able to find or understand them quickly. For example, Firsts was difficult to find and its name was relatively abstract, while some Weaning and health icons were not intuitive enough. These issues suggest that the existence of a feature does not necessarily mean that parents can understand what it is or where to find it.

Another issue concerns health and growth information. Content such as Growth Chart, percentiles, medical terms, and stool colour requires further interpretation before parents can judge their baby's condition. Therefore, the issue is not simply whether information is available, but whether the information can be understood and used for judgement.

In addition, the amount and text-heavy format of the NHS Guides can increase the burden of reading and filtering information. For practical tasks such as breastfeeding, burping, CPR, wrapping, bathing, and choking rescue, text alone may not be sufficient to support parents in carrying out the actions. Participants therefore suggested adding images, step-by-step instructions, and short videos.

Therefore, the core finding of this theme is that Bairn's features and content need to go through three stages—"discover, understand, and apply"—to provide practical value. Additional cognitive costs at any stage may reduce the practical value of the features.

6.3 Research Finding 1: Feature Discoverability Is Influenced by the Match Between Parents' Existing Parenting Concepts and the App's Information Architecture

Parents do not search for features randomly. They form expectations based on how they understand parenting tasks. For example, P4 expected Solid Food to be under Feeding, P5 initially could not find Weaning, and P4 found Firsts abstract and easy to overlook. Although these issues occurred in different features, they suggest that parents use existing parenting concepts to predict where a feature should be and what it means.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P4	Weaning does not match the expected category	"I would expect weaning to be under Feeding."
P5	Weaning feature is difficult to discover	"I couldn't find the Weaning feature at first."
P4	Firsts feature is difficult to find	"The entry is extremely hard to locate."
P4	Firsts has an abstract name	"The name 'Firsts' is too abstract."
P4	Firsts is easy to overlook	"This section is placed at the very bottom of the page, so it's easy to overlook."
P4	Earlier discovery could increase use	"If I had discovered the 'Firsts' milestone tab earlier, I would use it frequently."
P5	The "+" button is difficult for first-time users to notice	"The app is not very convenient to use. For example, the "+" button in the top-left corner is difficult for first-time users to notice."
P3	The "+" button is not intuitive for adding records	"First time I used the Growth section, I couldn't work out how to add a new record. I spent quite a while before realising that I had to tap the "+" button"
P5	Feature entry points are difficult to discover	"At first I could not find it, and even after I found it, it was not easy to use. There is no clear guidance to help users discover or understand the feature."

Interpretation:

The evidence suggests that feature discoverability is not simply a navigation issue. It is also related to whether the app's information architecture matches parents' mental models. Parents often look for a feature in response to a specific parenting task rather than systematically exploring the whole app. When categories or names do not match their understanding of concepts such as feeding, weaning, or milestones, they may need additional exploration to find the appropriate feature. Therefore, features become more discoverable when they are organised

and named in ways that correspond to parents' existing parenting concepts.

6.4 Research Finding 2: The Main Difficulty with Health and Growth Information Is Turning Professional Information into Judgement About the Baby's Condition

Issues with Growth Chart, percentiles, medical terminology, and stool colour show that after receiving health and growth information, parents still need to interpret it and judge what it means for their baby.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P2	Difficulty understanding the Growth Chart	"It took me a while to understand what the chart was showing."
P2	Difficulty understanding chart colours	"I wasn't sure what the different colours represented."
P2	Difficulty understanding percentiles	"I wasn't sure what the percentiles actually represented."
P2	Difficulty judging the baby's growth status	"I couldn't tell what stage of growth my baby was in."
P7	Difficulty understanding Growth Chart percentiles	"I didn't understand the growth chart percentile graph."
P7	Difficulty understanding medical terminology	"I didn't understand the medical terminology used in that section."
P7	Uncertainty about stool colour	"Sometimes my baby's stool changes colour, and I don't know whether it's normal or a sign of illness."
P8	Difficulty understanding text-based CPR information	"I find text-only CPR instructions difficult to understand."

Interpretation:

The deeper issue is that seeing information does not automatically mean that parents can use it. Parents need to move through a process of seeing information, understanding it, and judging what it means for their baby. For example, after viewing a Growth Chart, parents may still need to understand colours and percentiles and determine how these relate to their baby's development. Similarly, observing stool-colour changes does not by itself tell parents whether the change is normal or concerning. Therefore, health information needs to reduce the distance between professional information and everyday parental judgement, rather than simply providing more content.

6.5 Research Finding 3: The Amount and Presentation of Parenting Information Affect the Cost of Filtering and Understanding It

Participants valued parenting guidance, but large amounts of content, text-heavy presentation, limited visual support, and unclear organisation could increase the effort required to find and understand relevant information.

Supporting Evidence

Participant	Open Code	Supporting Evidence
P2	Large amount of content in NHS Guides	"There was a lot of content."
P2	Lack of images and videos	"It was all text, without any pictures or videos."
P2	NHS Guides are not intuitive	"It didn't feel very intuitive."
P2	Reduced continued browsing	"I didn't continue reading."
P3	Need for more illustrations	"They would be even better if they included more pictures and illustrations."
P3	Images support understanding	"Feeding positions would be much easier to understand with a few images instead of only text."
P8	Difficulty understanding text-only content	"I find text-only CPR instructions difficult to understand."
P8	Need for visual guidance	"I would like to see some diagrams for guidance."
P8	Need for short instructional videos	"I would like to see some short videos for practical guidance."

Interpretation:

The issue is not simply that parents prefer images or videos, but that information processing can be demanding. When there is a large amount of information, parents need to filter it themselves. When information is mainly text-based, they also need to translate the written content into their own understanding. This is particularly important for content related to practical parenting tasks. Images, illustrations, and short videos can therefore reduce the cognitive cost of moving from information to understanding. Organising information according to the baby's age or developmental stage could also reduce the effort required to identify what is most relevant at the current stage.

6.6 Research Finding 4: Specific parenting knowledge cannot always be directly translated into practical actions; therefore, parents need demonstrative and step-by-step support

The needs raised by P8 regarding breastfeeding positions, burping, CPR, wrapping, bathing, and choking rescue should be retained as a separate finding because they reveal an important distinction: knowing “what should be done” is not the same as knowing “how to do it in practice”. For new parents, some parenting information may be available, but this information still needs to be transformed into specific actions. Tasks such as feeding positions, burping, wrapping a baby, and emergency responses involve physical movements and sequences that are difficult to understand only through written descriptions. Therefore, parents do not only need access to knowledge; they need intermediate support that helps transform knowledge into action.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P8	Gap between parenting knowledge understanding and practical execution	"The NHS Guides are very good, but I think they would be even better if they included more pictures and illustrations. For example, feeding positions would be much easier to understand with a few images instead of only text."
P8	Physical parenting actions require visual demonstrations	"Feeding positions would be much easier to understand with a few images instead of only text."
P8	Practical parenting skills require step-by-step guidance	"All key actions (including breastfeeding positions, burping, baby CPR positions, wrapping, bathing, umbilical cord care, choking rescue) should include short videos and step-by-step illustrations."
P8	Practical skills require video demonstrations	"I think practical skill videos would be helpful."
P8	High-risk operations cannot rely on text-only explanations	"I wanted to understand the standard compression and artificial respiration operations, but the page did not provide diagrams or short videos. It was difficult to understand through text only, and I didn't know whether I was doing it correctly."
P8	CPR and choking rescue require direct demonstrations	"All key actions ... should include short videos and step-by-step illustrations."

Interpretation:

This finding further develops Theme 2's finding that parents need parenting guidance. Theme 2 explains why parents seek knowledge and guidance; Theme 5 further explains why existing knowledge may still be insufficient. For practical parenting tasks, written knowledge needs to be transformed into concrete actions. Parents therefore do not simply lack knowledge; they lack

intermediate support that helps convert knowledge into behaviour.

The value of demonstration videos, illustrations, and step-by-step guidance is that they reduce the conversion cost between “knowledge → action”. This logic provides a stronger research finding than simply stating that “users want videos”.

6.7 Research Finding 5: Continuous use can help parents gradually adapt to the App, but user adaptation cannot replace low-learning-cost design

Some participants reported that after using the app for a period of time, they became more familiar with feature locations, improved recording efficiency, and experienced smoother interactions. This finding should not be treated as an independent positive theme, but rather as a process explanation of previous usability problems.

Although users can adapt through repeated use, this adaptation also indicates that the initial design requires users to invest additional time and effort to understand the system. A better design should reduce the need for exploration and trial-and-error instead of relying on users to gradually become familiar with the app.

Supporting Evidence:

Participant	Open Code	Supporting Evidence
P2	Initial exploration and learning needs	"When I first started using the app, I wanted to learn more about parenting and keep a good record of my child's growth."
P2	Lack of guidance affects continued use	"Many of the features lack helpful guidance and are inconvenient to use, which has reduced how often I use the app."
P3	Repeated use creates daily operation routines	"When my baby wakes up, I first check whether the nappy needs changing. Then I open the app, stop the Sleep recording, compare my baby's sleep with the previous day, record the nappy change, and finally record the feeding."
P6	Initial difficulty understanding feature meaning	"When I first opened it, it took me a while to understand what the chart was showing."
P6	Feature location increases initial learning cost	"Some features were buried quite deeply in the app, so it took me some time to find them when I first started using it."

Interpretation

Users can indeed adapt to the app through repeated use, but becoming familiar with the app does not prove that the initial design has no problems. Instead, the adaptation process shows that parents need to invest time to build an understanding of the system.

Considering Theme 3's finding that parents have limited time and energy in daily childcare, this

additional learning cost is meaningful. Better design should minimise exploration costs during first-time use rather than depending entirely on users' own trial and error.

6.8 Theme-level Interpretation

Combining the above findings, some current problems in Bairn can be understood as three continuous barriers.

The first is the "discovery barrier": parents need to find features and understand what they are. Difficulties finding Firsts, abstract naming of Firsts, unclear weaning icons, and unclear health icons belong to this level.

The second is the "interpretation barrier": after finding information, parents still need to understand what it means. Problems with Growth Chart, percentiles, medical terminology, and stool colour show that professional information cannot automatically become usable judgement without clear explanation.

The third is the "action conversion barrier": even when parents understand parenting knowledge, they may still not know how to perform specific actions. P8's requests for demonstrations of breastfeeding, burping, CPR, wrapping, bathing, and choking rescue show that practical parenting tasks require more step-by-step and visual support.

Therefore, the core logic of this theme can be summarised as: Discovery of features → Understanding information → Conversion into action. Any additional cognitive cost introduced at these stages may weaken the practical value generated by the app.